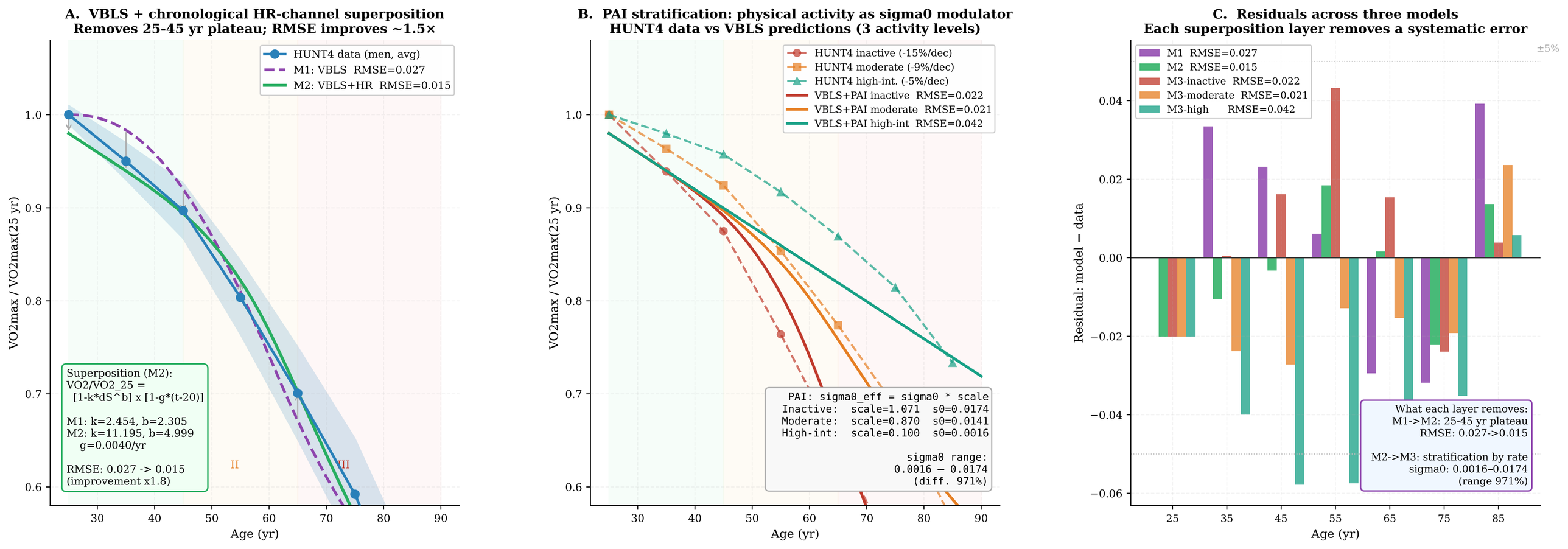

A. VBLS + chronological HR-channel superposition
Removes 25-45 yr plateau; RMSE improves ~1.5×
HUNT4 data (men, avg)
M1: VBLS RMSE=0.027
M2: VBLS+HR RMSE=0.015
VO2max / VO2max(25 yr)
Age (yr)
Superposition (M2):
VO2/VO2_25 =
[1-k*dS^b] x [1-g*(t-20)]
M1: k=2.454, b=2.305
M2: k=11.195, b=4.999
g=0.0040/yr
RMSE: 0.027 -> 0.015
(improvement x1.8)
II
III
B. PAI stratification: physical activity as sigma0 modulator
HUNT4 data vs VBLS predictions (3 activity levels)
HUNT4 inactive (-15%/dec)
HUNT4 moderate (-9%/dec)
HUNT4 high-int. (-5%/dec)
VBLS+PAI inactive RMSE=0.022
VBLS+PAI moderate RMSE=0.021
VBLS+PAI high-int RMSE=0.042
PAI: sigma0_eff = sigma0 * scale
Inactive: scale=1.071 s0=0.0174
Moderate: scale=0.870 s0=0.0141
High-int: scale=0.100 s0=0.0016
sigma0 range:
0.0016 – 0.0174
(diff. 971%)
C. Residuals across three models
Each superposition layer removes a systematic error
M1 RMSE=0.027
M2 RMSE=0.015
M3-inactive RMSE=0.022
M3-moderate RMSE=0.021
M3-high RMSE=0.042
±5%
Residual: model − data
What each layer removes:
M1->M2: 25-45 yr plateau
RMSE: 0.027->0.015
M2->M3: stratification by rate
sigma0: 0.0016–0.0174
(range 971%)